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SRI RAMAKRISHNA ENGINEERING COLLEGE, COIMBATORE

(Autonomous Institution, Reaccredited by NAAC with 'A+' Grade Approved by AICTE and Permanently affiliated to Anna University, Chennai)

AUTONOMOUS EXAMINATIONS - MAY'2025

COMMON TO B.E – CSE / B.TECH – IT & M.TECH - CSE

20MA203 - LINEAR ALGEBRA AND DIFFERENTIAL CALCULUS

Answer ALL questions

Duration: 3 Hours

Maximum: 100 Marks

PART - A

(7x 1 = 7 Marks)

- CO1** Select the suitable expression for diagonalisation of a matrix A.
a) $D = N A N^T$ b) $D = N^T A N$ c) $D = A^T N A$ d) $D = A^T N A^T$
- CO1** Infer the value of 'k' in the characteristic equation $\lambda^3 - 3\lambda + k = 0$ of the matrix
 $A = \begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix}$ is _____.
a) 0 b) 1 c) 2 d) -2
- CO2** The vector space V is said to be the direct sum of its subspaces U and W, then _____.
a) $U \cup W = V$ b) $U \cap W = \phi$ c) $V = U \oplus W$ d) $U \subset W$
- CO2** The dimension of the null space of a matrix A is known as _____.
a) rank b) nullity c) span d) determinant
- CO3** A function $T : V \rightarrow V$ is called _____ transformation if $T(x) = x$ for all $x \in V$.
a) Zero b) Linear c) Identity d) one to one
- CO4** Select the radius of curvature for the curve $y = e^x$ at the point (0,1)
a) $\sqrt{2}$ b) 0 c) $2\sqrt{2}$ d) $\frac{1}{\sqrt{2}}$

7. **CO5** If $u = f(x, y)$ is an implicit function in x and y then $\frac{dy}{dx} = \underline{\hspace{2cm}}$.

- a) $\frac{f_x}{f_y}$ b) $\frac{f_y}{f_x}$ c) $-\frac{f_x}{f_y}$ d) $-\frac{f_y}{f_x}$

Fill in the blanks:

(3x 1 = 3 Marks)

8. **CO2** The locus of centre of curvature is called .

9. **CO5** If u and v are functions of x and y then $\frac{\partial(u,v)}{\partial(x,y)} \times \frac{\partial(x,y)}{\partial(u,v)} = \underline{\hspace{2cm}}$.

10. **CO2** The curvature of a curve is equal to .

PART - B

(5x 3 = 15 Marks)

11. **CO1** Find the Characteristic equation of the matrix $\begin{pmatrix} 1 & 2 \\ 0 & 2 \end{pmatrix}$

12. **CO2** Let $V = \mathbb{R}^3$, show that $W = \{ (x, y, z), x = z + 2 \}$ is not a subspace of V .

13. **CO3** Show that f is Linear $f : \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = 2x$

14. **CO4** Find the curvature of the curve $2x + 3y = 4$.

15. **CO5** If $u = (x-y)(y-z)(z-x)$ show that $\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z} = 0$

PART - C

(5 x 15 = 75 Marks)

Q.No: 16 Compulsory Question

Q.No: 17 to 22 Answer any FOUR Questions

Compulsory Question:

16. **CO1** Reduce the quadratic form $6x_1^2 + 3x_2^2 + 3x_3^2 - 4x_1x_2 - 2x_1x_3 + 4x_2x_3$ to (15)
canonical form by an orthogonal transformation. Also find rank, index,
signature and nature of the quadratic form.

17. **CO1** i) Find the Eigen values and Eigen vectors of the matrix (8)

$$A = \begin{bmatrix} 7 & -2 & 0 \\ -2 & 6 & -2 \\ 0 & -2 & 5 \end{bmatrix}$$

CO5 ii) A rectangular box open at the top is to have a volume of 32cc. Find the (7)
dimensions of the box, that requires the least material for its construction.

18. **CO2** i) Prove that $\mathbb{R}^2$ with usual addition and scalar multiplication is a vector space. (8)

- CO2** ii) Find bases for the row space and column space of the following matrix (7)

$$A = \begin{pmatrix} 1 & 1 & 4 & 1 & 2 \\ 0 & 1 & 2 & 1 & 1 \\ 0 & 0 & 0 & 1 & 2 \\ 1 & -1 & 0 & 0 & 2 \\ 2 & 1 & 6 & 0 & 1 \end{pmatrix}$$

19. **CO3** i) Let A be the matrix on $\mathbb{R}^2$ and basis S of $\mathbb{R}^2$ be $A = \begin{bmatrix} 3 & -2 \\ 4 & -5 \end{bmatrix}$ and (8)

$S = \{u_1, u_2\} = \{(1, 2), (2, 5)\}$. Find the matrix representation with respect to a) basis b) usual basis.

- CO3** ii) Consider the following two bases of $\mathbb{R}^3$. (7)

$$E = \{e_1, e_2, e_3\} = \{(1, 0, 0), (0, 1, 0), (0, 0, 1)\}$$

$$\text{and } S = \{u_1, u_2, u_3\} = \{(1, 0, 1), (2, 1, 2), (1, 2, 2)\}$$

- Find the change of basis matrix P from basis E to the basis S.
- Find the change of basis matrix Q from S to the basis E.

20. **CO4** i) Find the radius of curvature at the point $\left(\frac{3a}{2}, \frac{3a}{2}\right)$ for the curve (8)

$$x^3 + y^3 = 3axy$$

- CO4** ii) Find the equation of the evolute of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ (7)

21. **CO4** i) Find the centre and circle of curvature of the curve $xy = c^2$ at (c, c) . (8)

- CO5** ii) If $z = f(u, v)$ and $u = x^2 - y^2, v = 2xy$, show that (7)

$$\left(\frac{\partial z}{\partial x}\right)^2 + \left(\frac{\partial z}{\partial y}\right)^2 = 4(u^2 + v^2)^{\frac{1}{2}} \left[\left(\frac{\partial z}{\partial u}\right)^2 + \left(\frac{\partial z}{\partial v}\right)^2 \right]$$

22. **CO5** i) Find the Jacobians of y_1, y_2, y_3 with respect to, (8)

x_1, x_2, x_3 if

$$y_1 = \frac{x_2 x_3}{x_1}, y_2 = \frac{x_1 x_3}{x_2}, y_3 = \frac{x_2 x_1}{x_3}$$

CO5 ii) Expand $e^x \cos y$ in powers of x and y as far as the terms of the third (7)
degree.
